

Customer Sales Analysis

Week 5 Project — Advanced Data Manipulation with Pandas

Total Revenue	\$12,365,048
Total Orders	100
Average Order Value	\$123,650
Top Customer	CUST016 — \$373,932
Customers in Relationship Data	500
Overall Churn Rate	10.6%

Data sources: **sales_data.csv** (100 orders, Jan–Apr 2024) merged with **customer_data.csv** (500 customer relationship / churn records).

1. Project Overview

This report analyzes customer purchasing patterns using two datasets: a sales transaction log and a customer relationship/churn file. The goal is to identify top customers, understand regional and product-level sales performance, surface seasonal trends, and turn churn data into concrete retention recommendations. The full, reproducible analysis lives in **customer_analysis.ipynb**.

2. Data & Methodology

sales_data.csv has 100 rows: Date, Product, Quantity, Price, Customer_ID, Region, Total_Sales.

customer_data.csv has 500 rows: CustomerID, Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitizen, Churn. Neither file had missing values or duplicate rows.

Merging note: the two files use different ID formats (*CUST001* vs. *C00001*) and were not built with a shared key. To demonstrate a real merge, IDs were aligned by numeric suffix (sales customer #1 ↔ relationship record #1). This is a documented assumption for this exercise, not a claim of a verified shared identity system.

Techniques applied: datetime feature extraction (year/month/day/weekday), string cleaning (*.str.strip().str.title()*), multi-condition filtering (AND/OR), an inner merge, groupby aggregations (sum, mean, count), and two pivot tables (Region × Product, Region × Month).

3. Key Findings

Who are our most valuable customers?

The top 10 customers by single-order value are led by Laptop and Tablet buyers, topped by CUST016 at \$373,932. Adding relationship-side spend (TotalCharges) as a lifetime-value proxy shifts the ranking slightly toward longer-tenured customers — a reminder that one large order and a valuable long-term relationship aren't always the same customer.

What sells best, and what sells together?

Laptop is the strongest individual product (\$3.89M), followed closely by Tablet (\$2.88M) and Phone (\$2.86M); Headphones and Monitor trail well behind. A true cross-sell / market-basket analysis isn't supportable here: each of the 100 customers has exactly one recorded order, so there's no repeat-purchase history to mine. As a real, supportable substitute, product mix was compared by region instead (see notebook) — useful for regional stocking decisions even without basket data.

Which regions have the highest sales?

North (\$3.98M) and South (\$3.74M) lead, together making up the majority of revenue; East (\$2.52M) and West (\$2.12M) trail. Product mix is not uniform across regions, so inventory should follow the region × product pivot table rather than assuming every region needs the same stock.

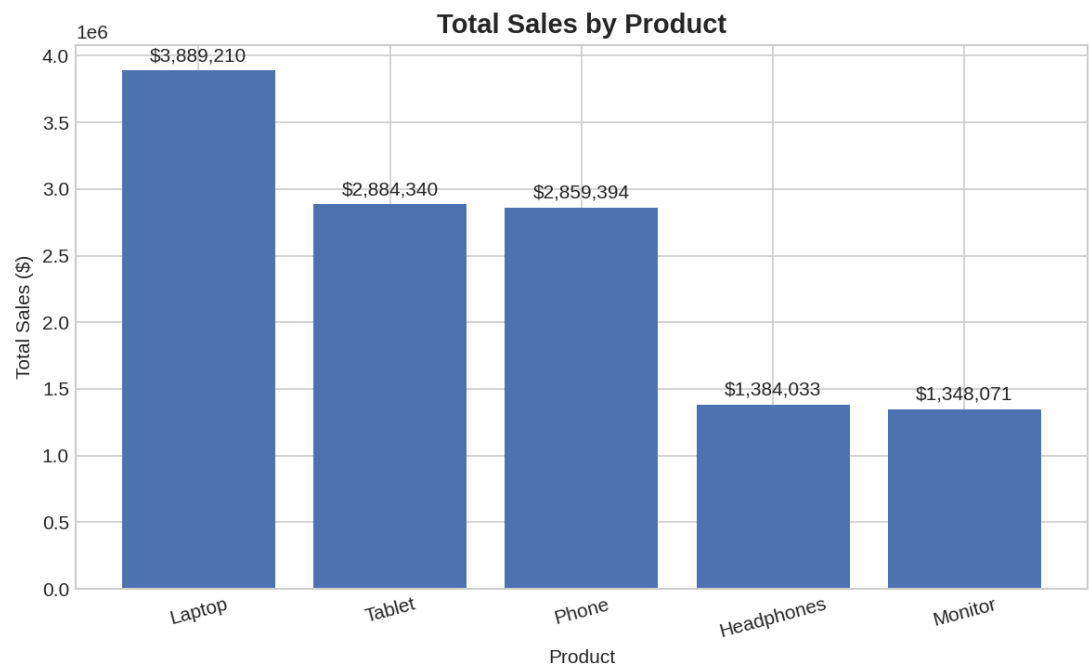
What are the seasonal trends?

Monthly sales rose from January into a March peak (\$4.49M), then the recorded total for the following month is much lower. That final month in the dataset is partial (data ends April 9), so the drop reflects fewer days recorded, not a real downturn.

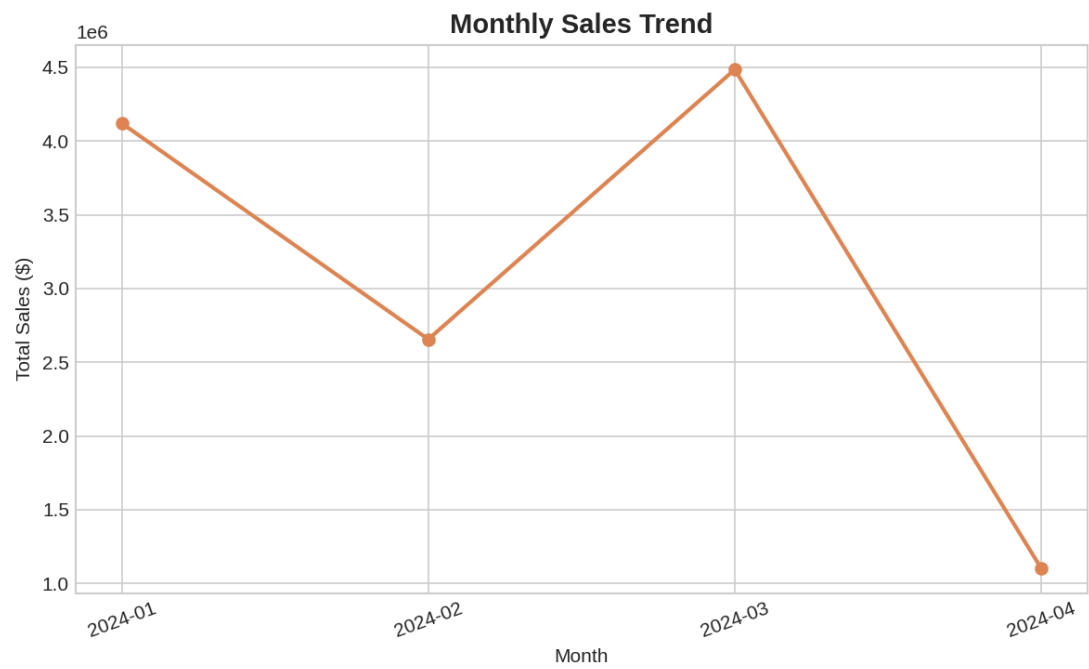
How can we improve customer retention?

Churn is heavily concentrated in month-to-month contracts (20.6%) versus two-year (6.9%) and one-year (4.3%) contracts. Overall churn across all 500 relationship records is 10.6%. The clearest lever is migrating high-value, month-to-month customers onto longer contracts.

4. Dashboard

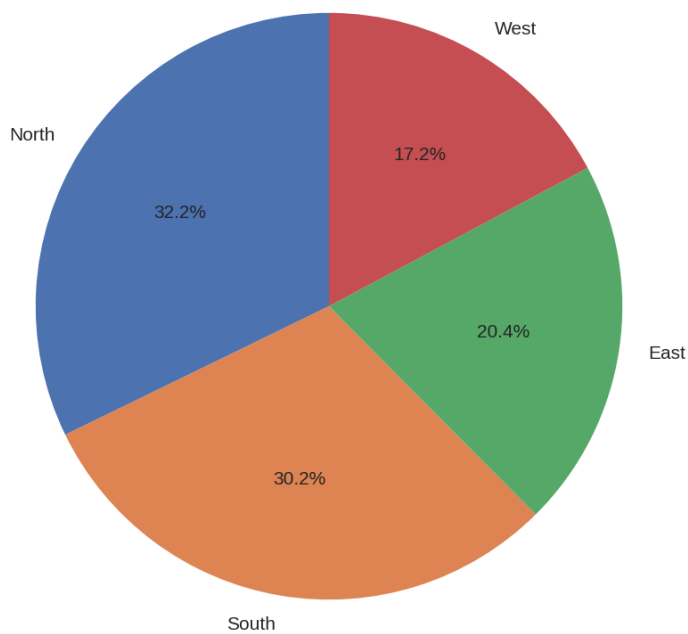


Total Sales by Product



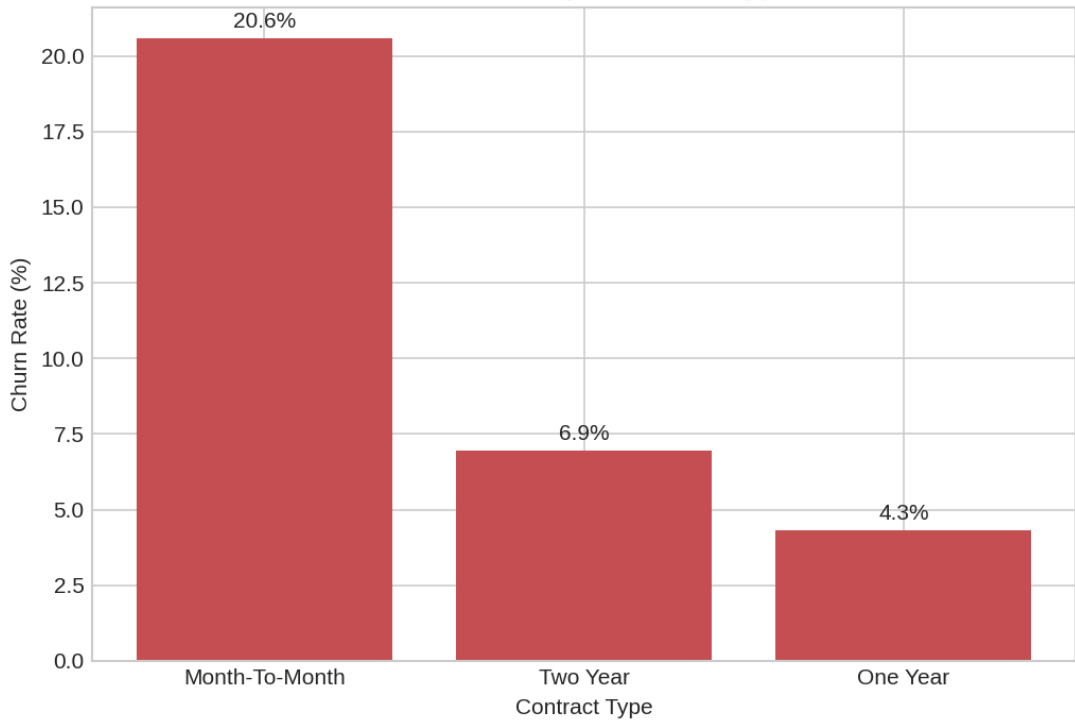
Monthly Sales Trend

Sales Share by Region

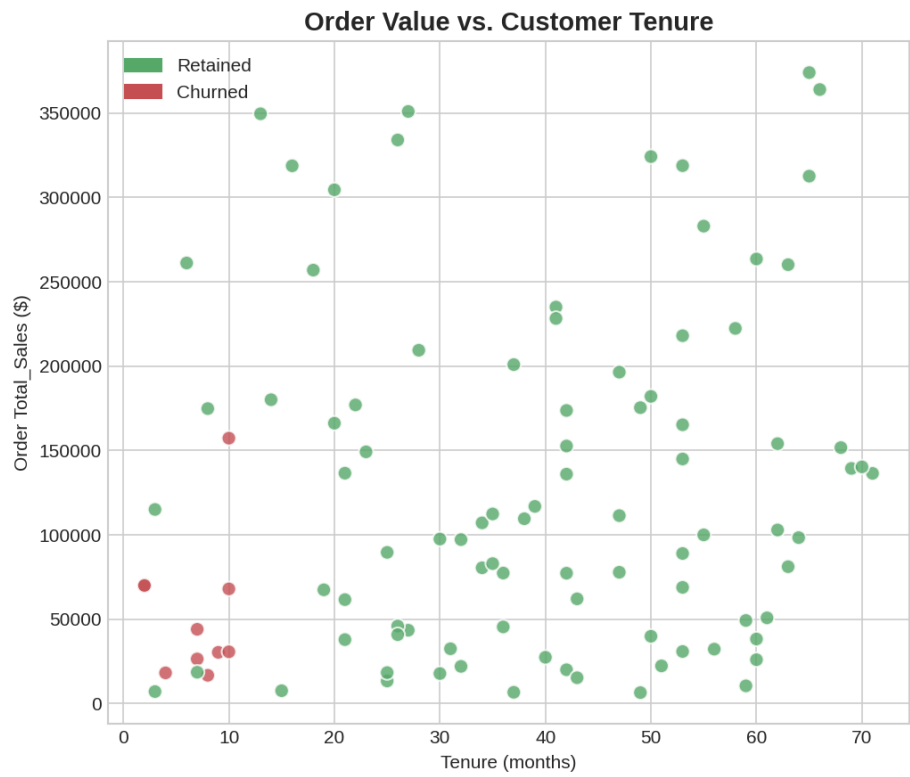


Sales Share by Region

Churn Rate by Contract Type



Churn Rate by Contract Type



Order Value vs. Customer Tenure (colored by churn status)

5. Business Recommendations

1. Prioritize win-back outreach to customers who are both high-value (top quartile of order value) and already churned — they generated strong revenue before leaving and are the most valuable to recover.
2. Pilot an incentive (discount or added perk) for month-to-month customers to switch to an annual contract, starting with customers above the 75th-percentile spend, since that segment shows the highest churn risk relative to its value.
3. Align regional inventory and marketing with the Region × Product sales pivot rather than stocking every region identically.
4. Re-run this analysis once repeat-purchase data exists so a genuine cross-sell / market-basket analysis becomes possible — the current one-order-per-customer structure can't support it.

Full reproducible code, all intermediate tables, and additional detail are in **customer_analysis.ipynb**.